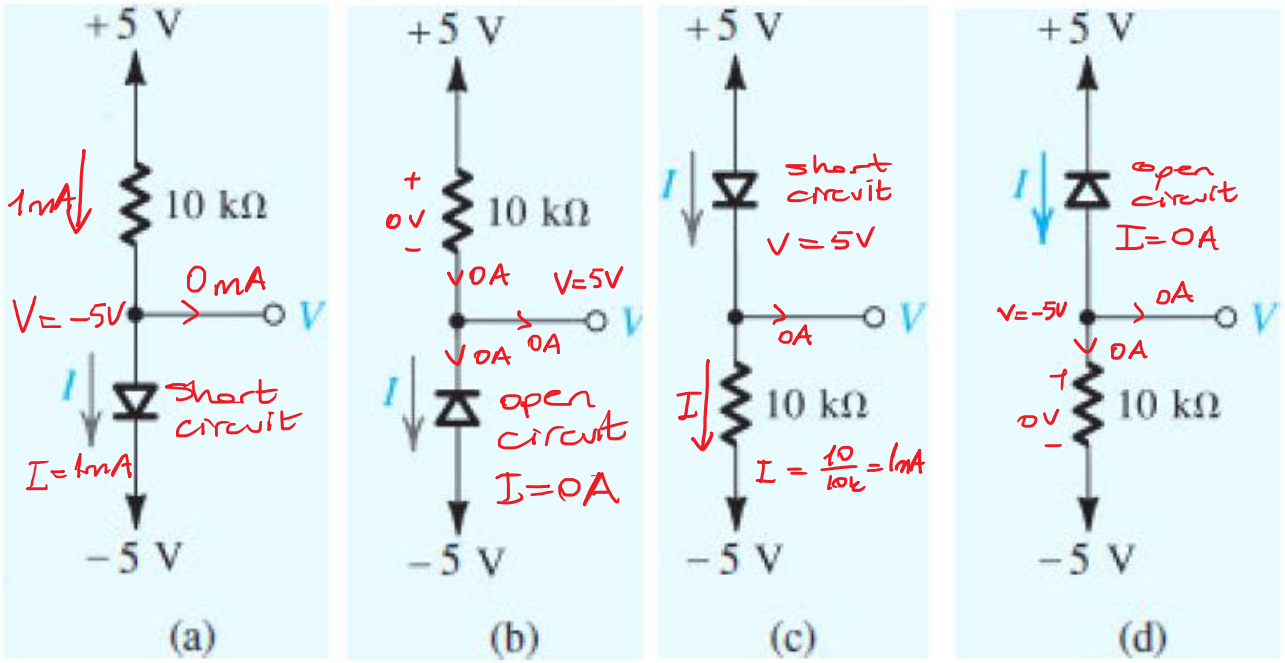
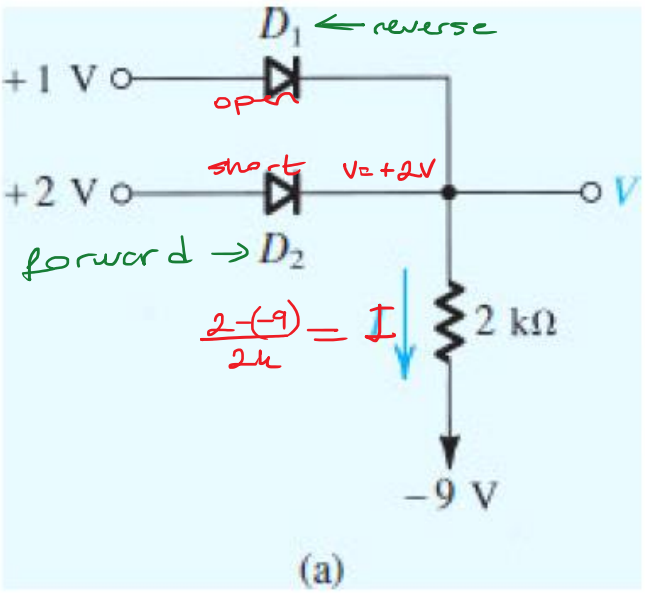
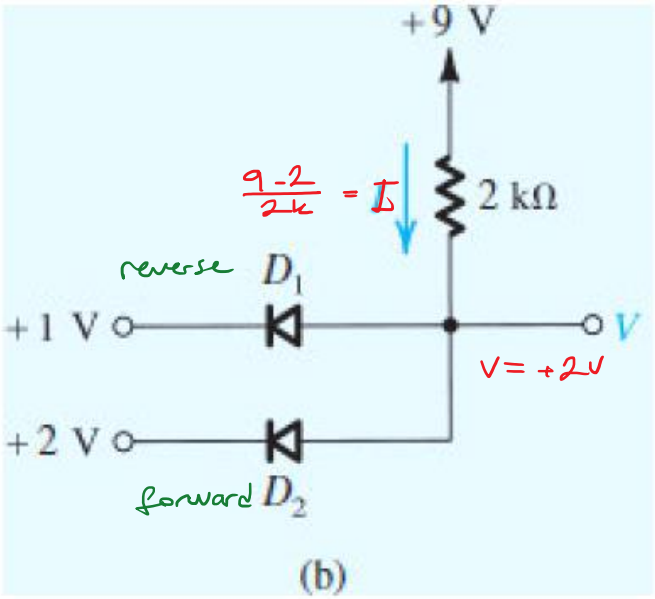
P4.2



forward reverse forward reverse

P4.3

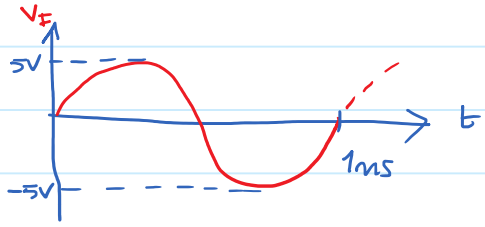




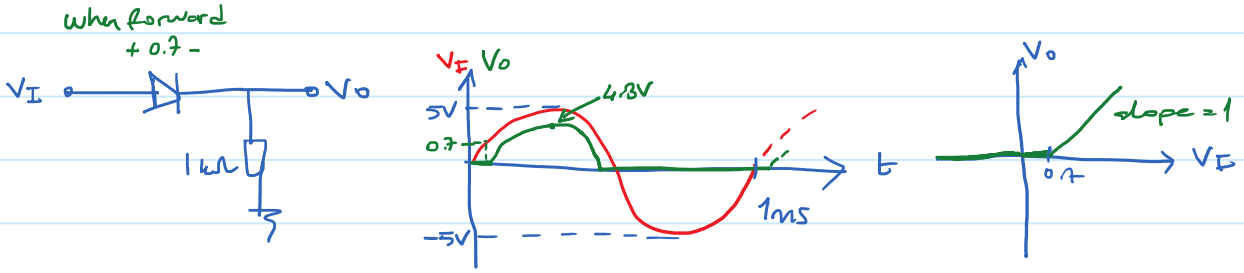
	D1	D2
	forward	forward
	forward	reverse
valid	reverse	forward
	reverse	reverse

	D1	D2
	forward	forward
	forward	reverse
valid consistent with the model	reverse	forward
	reverse	reverse

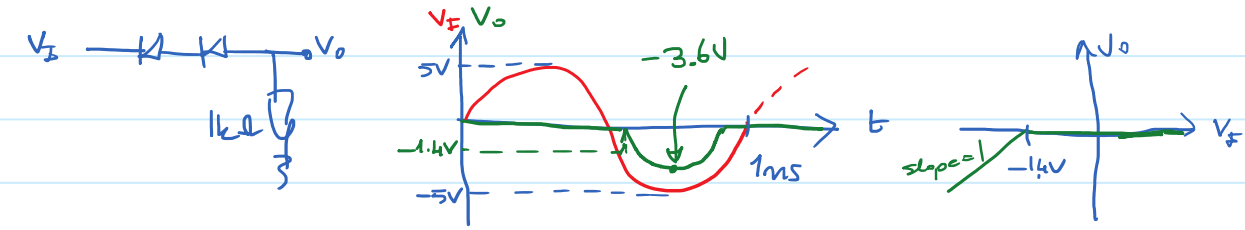
P4.4



a)



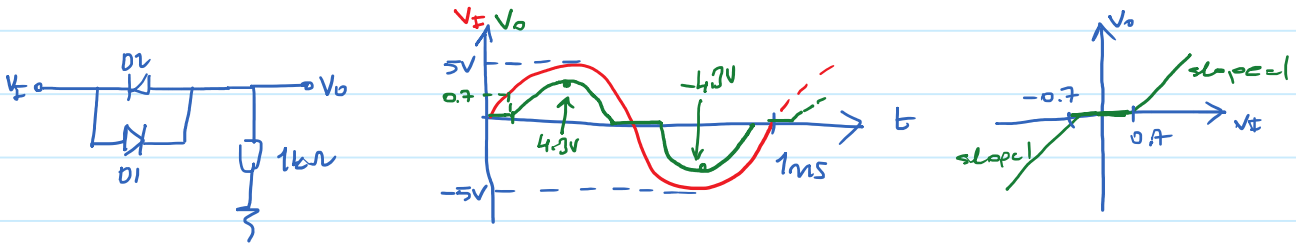
b)



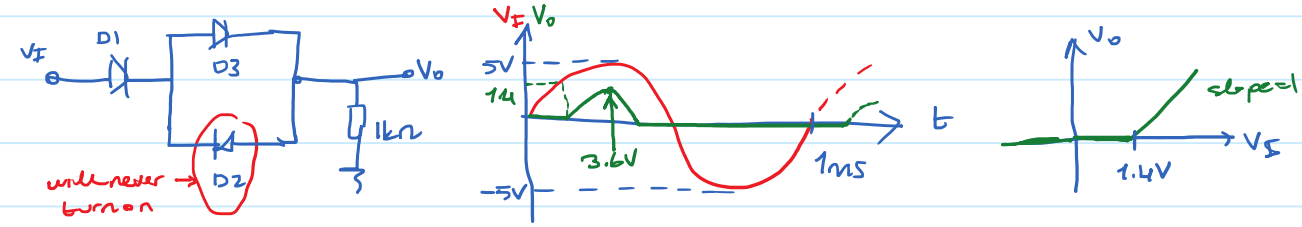
c) either D1 or D2 will be cut-off (reverse biased) $V_o = 0$ always

d) same as (a)

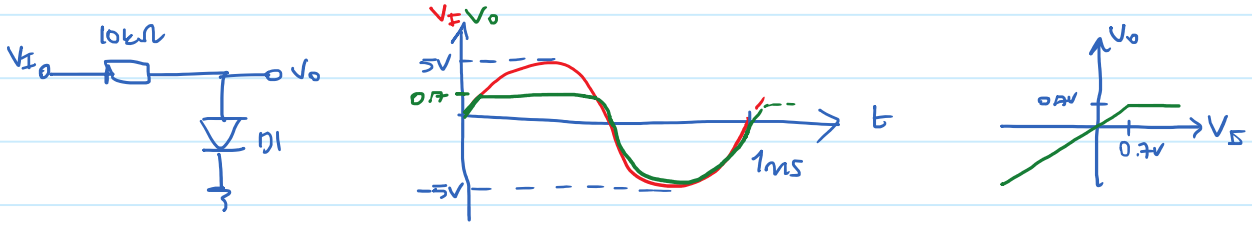
e)



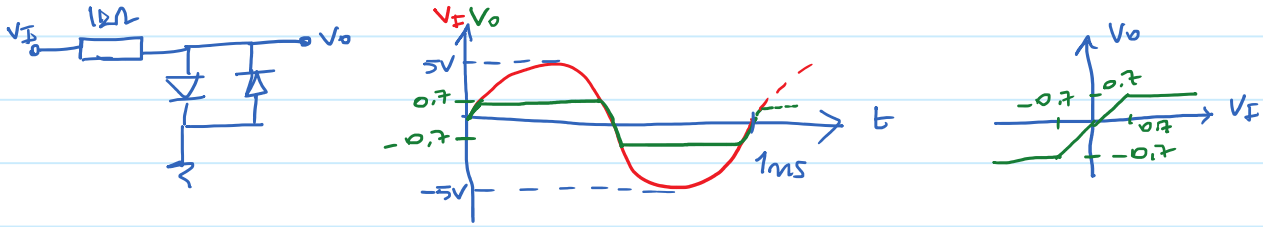
f)



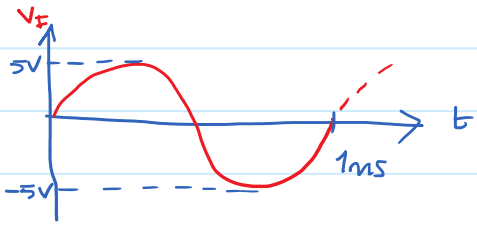
g)



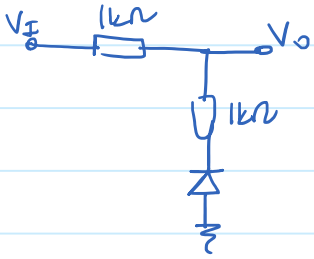
h)

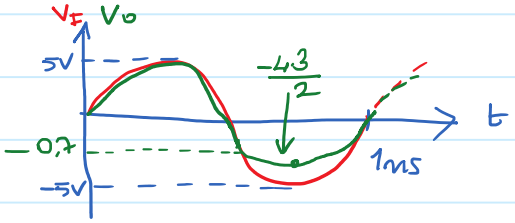


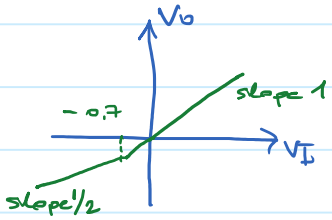
P4.4 ...



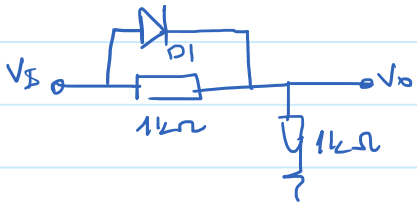
i)

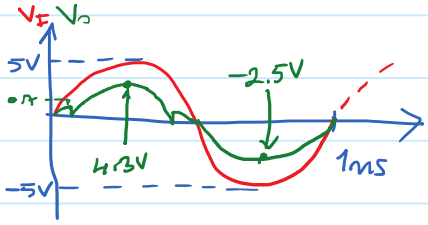


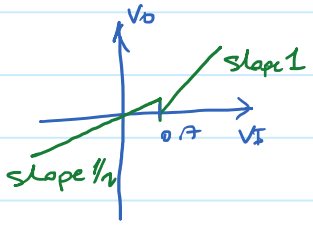




j)

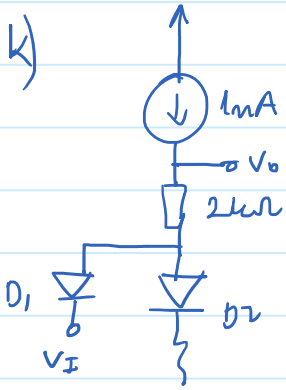


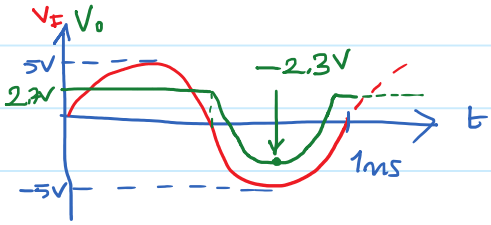


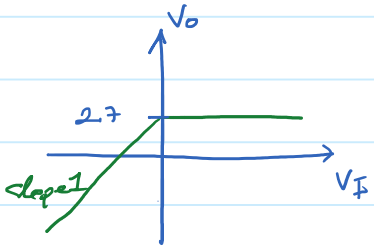


in reality not discont.
this is due to simplified model

k)







P4.18

$$V_T = \frac{kT}{q} = \frac{1.38 \times 10^{-23} \cdot T}{1.60 \times 10^{-19}}$$

C	−55	0	40	125
K	218	273	313	398
V_T	18.8m	23.5m	27m	34.3mV

$$25m = \frac{k\,T(^\circ K)}{q} \Rightarrow T(^\circ C) = 16.85^\circ C$$

P4.20

forward mode $\Rightarrow I \cong I_S e^{v/V_T}$

$$\frac{1\,m}{I} = \frac{I_S e^{0.7/V_T}}{I_S e^{0.6/V_T}}$$

$25\,mV$

$$\frac{1\,m}{I} = e^{(0.7-0.6)/V_T}$$

$$I = 1{,}8 \times 10^{-5}\,A = 18\,\mu A$$